

KENYA JUNIOR SCHOOL EDUCATION ASSESSMENT

GRADE 9 2025



MATHEMATICS

Time: 1 hour 30 Minutes

Name:

School:

Assessment Number:

Signature: Date:

Instructions for Learners

1. Write down your name, your school name and assessment number in the spaces provided.
2. This paper consists of **29 questions**.
3. Check the question paper to confirm that all the pages are printed and no questions are missing.
4. Answer all the questions in the spaces provided.

Score Grid

Question	Score	Question	Score
1-20		26.	
21.		27.	
22.		28.	
23.		29.	
24.		30.	
25.		31.	
		32.	
Total score		Assessed out of 50	

ASSESSMENT RUBRICS

Learning Area	Exceeds Expectations (4)	Meets Expectations (3)	Approaching Expectations (2)	Below Expectations (1)
Mathematics				

Section A (20 marks)

1. What is the value of the following?

$$\frac{22 - (-14)}{6 \times (-2)} - \frac{4^2 \times (-6) - 12}{72 \div (-8) \times 3}$$

A. -5 B. -3 C. -7 D. 4

2. Use mathematical tables to evaluate:

$$0.45^3 + \sqrt[3]{4394}$$

A. 16.4700986 B. 16.47
C. 0.09113 D. 16.46113

3. The volume of a cubic gift box is **0.729 m³**.
Calculate the length of one side of the cubic box.

A. 0.9 m B. 9 m C. 0.3874 m D. 38 m

4. A farmer harvested 144 bags of maize in one season. In the second season, the yield increased in the ratio 4:3. The farmer supplied all the bags of maize harvested in the second season equally to three millers. How many bags of maize did each miller get?

A. 192 B. 48 C. 64 D. 36

5. James simplified an algebraic expression and got the answer as x^4 . Which of the following expressions did he likely simplify?

A. $x^4 \div x^8$ B. $(x^2)^2$ C. $x^3 \times x^2$ D. $x^6 - x^2$

6. A tap can take 3 hours to fill a tank whose capacity is 6 000 litres. How long would it take the same tap to fill a tank whose capacity is 10 000 litres?

A. 1 h 40 min B. 6 h

C. 4 h D. 5 h

WORKING SPACE

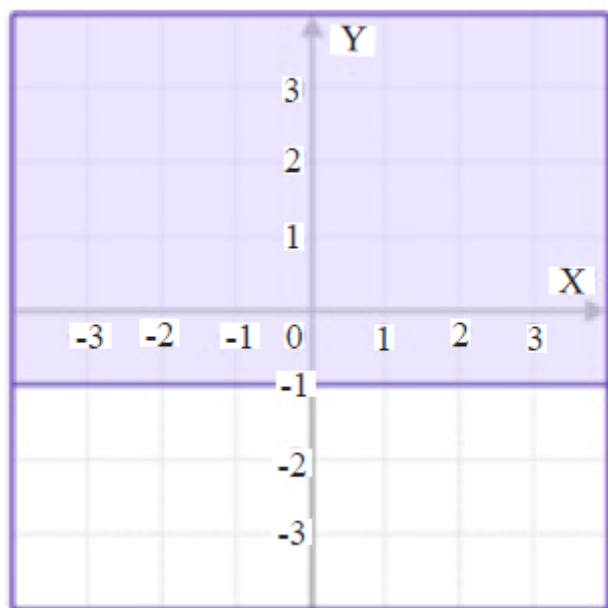
7. A class recorded the number of boys and girls participating in three different sports using the following matrix:

$$\begin{bmatrix} -1 & 5 \\ 2 & 1 \\ 4 & 0 \end{bmatrix}$$

What is the order of the matrix?

- A. 3×3 B. 3×2 C. 2×3 D. 2×2

8. Which of the following inequalities is represented by the shaded region?

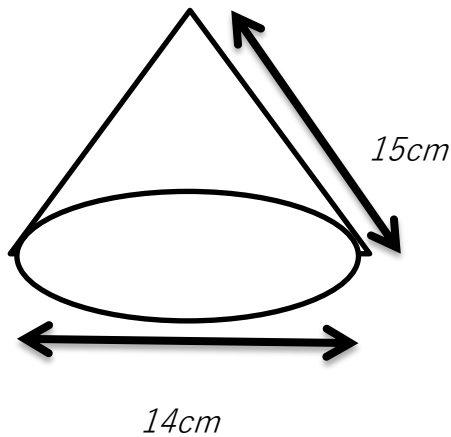


- A. $y \geq -1$ B. $y < 1$
C. $x \leq 1$ D. $x > -1$

9. A line which joins the points A (3,K) and B (− 2,5) has a gradient of $-\frac{2}{5}$. Find the value of K.

- A. 4 B. 5
C. 3 D. −2

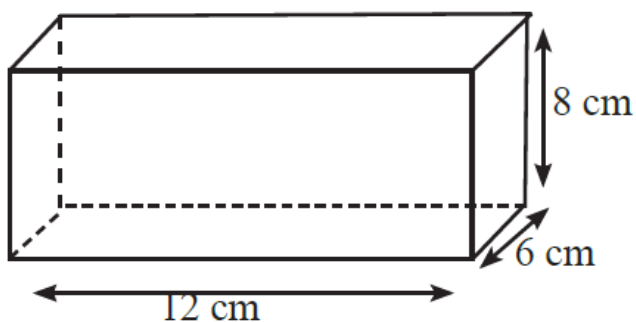
- 10.** An artist is making cone-shaped paper hats for a school play. Each hat has a diameter of **14 cm** and a slant height of **15 cm**.



How much paper is needed to make one hat?

- A. 330 cm^2 B. 220 cm^2
C. 154 cm^2 D. 484 cm^2
- 11.** A metal block has a mass of **12 000 kg** and a volume of **15 m^3** . What is its density?
- A. 750 kg/m^3 B. 800 kg/m^3
C. 850 kg/m^3 D. 900 kg/m^3

- 12.** The figure below represents an open rectangular container.



What is its total surface area?

- A. 96 cm^2 B. 360 cm^2
C. 432 cm^2 D. 576 cm^2

WORKING SPACE

13. Jacob drives **80 km in 2 hours**, then drives a further **60 km in 1.5 hours**. What is his average speed for the entire journey?

A. 40 km/h B. 50 km/h

C. 60 km/h D. 64 km/h

14. A car's velocity changes from **20 m/s to 50 m/s** in **5 seconds**. What is its acceleration?

A. 4 m/s² B. 6 m/s² C. 8 m/s² D. 10 m/s²

15. The cash price of a radio is **Sh 8 000**. The hire purchase price is **50% more** than the cash price. Jamila bought the radio on hire purchase by paying a deposit of **Sh 2 400** and equal monthly installments of **Sh 800**. In how many months did she pay the installments?

A. 18 B. 15 C. 12 D. 10

16. A school globe is spherical in shape with a radius of **14 cm**. What is its surface area?

$$\left(Take \pi = \frac{22}{7} \right)$$

A. 3 080 cm²

B. 2 464 cm²

C. 1 540 cm²

D. 1 760 cm²

17. Which quadrilateral has all the following properties?

i. *Opposite sides are equal*

ii. *Interior angles add up to 360°*

iii. *Some angles are acute while some are obtuse*

iv. *Diagonals bisect at right angles*

A. Parallelogram B. Rectangle

C. Rhombus D. Square

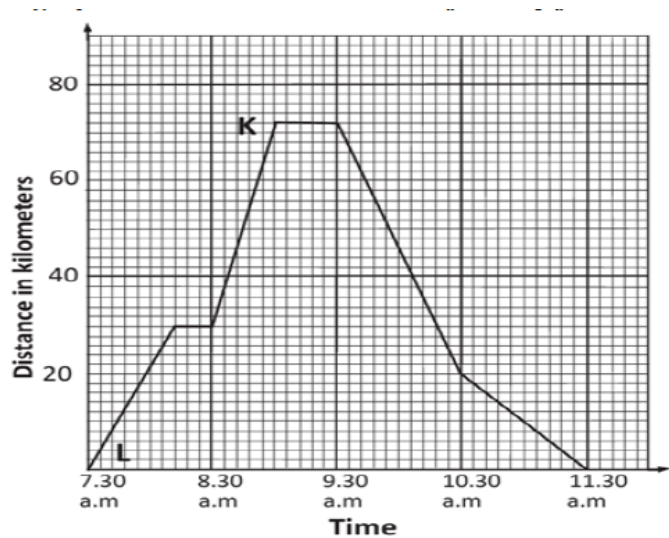
18. The figure below represents a rectangular plot of land drawn to scale **1:1000**.



What is the actual area of the plot of land in hectares?
A. 2 400 B. 24 C. 2.4 D. 0.24

19. Calculate the number of sides of a regular polygon whose exterior angle is **120°** .
A. 6 B. 3 C. 4 D. 5

20. The graph below shows a motorist’s journey from town L to town K and back.



What was the motorist’s average speed in km/h for the whole journey?
A. 36 km/h B. 18 km/h
C. $37\frac{17}{19}$ km/h D. 35 km/h

WORKING SPACE

Section B (30 marks)

21. Write the following expressions in logarithmic notation. (2 marks)

$$10^4 = 10\,000$$

22. Evaluate: (3 marks)

$$\frac{1}{2} \text{ of } \left(3\frac{1}{3} \times \frac{2}{5} \right) + 4\frac{3}{4} \div \frac{1}{4} ?$$

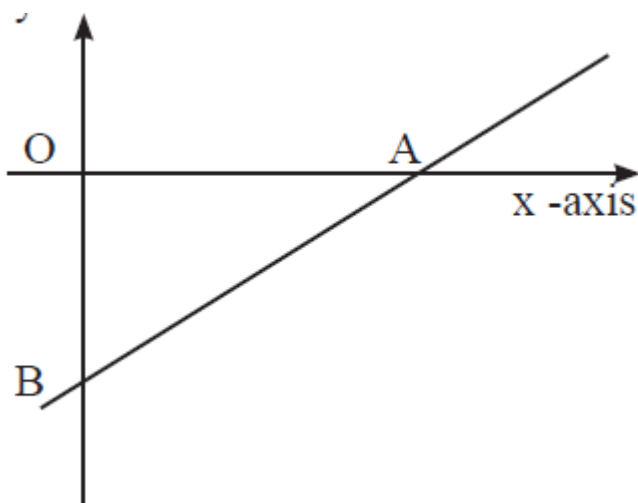
23. 4 men can paint a wall in **8 days**. How long will it take **8 men** to paint the same wall working at the same rate? (2 marks)

24. A rectangular garden has its length as $(2x+4)$ m and width as $(x-2)$ m. If the perimeter of the garden is less than **40 m**:

(a) Form a linear inequality to represent this information. (2 marks)

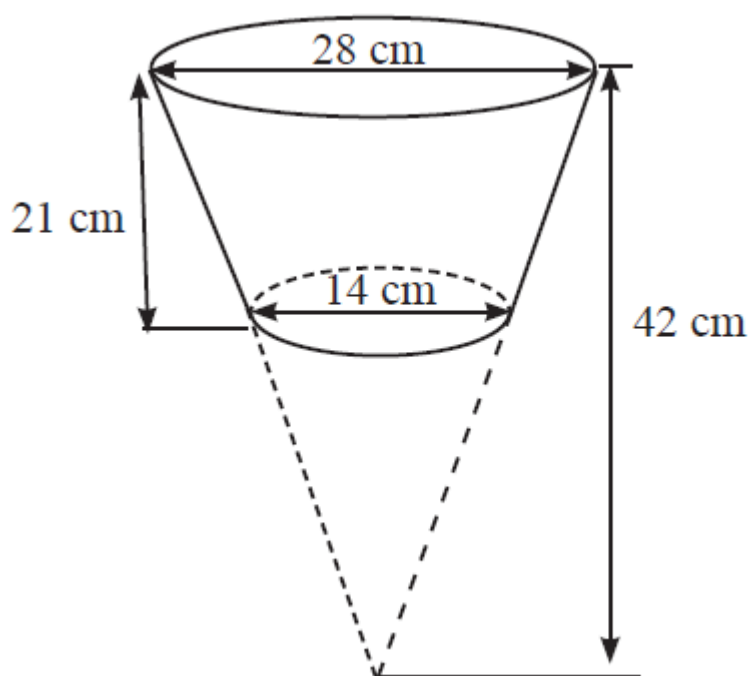
(b) Solve for x in the inequality. (2 marks)

25. On the diagram below, the line whose equation is $2y+3x=12$ passes through the points A and B. Determine the coordinates of points A and B. (3 marks)



26. Ngilu borrowed **Ksh 60 000** from a bank. The bank charged a simple interest at the rate of **10% per annum**. She paid back the money after a period of **two years**. Find the amount she paid back. (3 marks)

27. A flowerpot is shaped like a frustum of a cone with a **top diameter of 28 cm**, a **bottom diameter of 14 cm**, and **height 21 cm** as shown below. Determine the volume of the flowerpot. (Take $\pi = \frac{22}{7}$) (3 marks)



28. Draw a regular pentagon ABCDE of length **2.5 cm**. (4 marks)

29. A basket contains balls numbered from **30 to 40**. A student randomly picks one ball. Find the probability that the number on the ball is:

(a) a prime number. (2 marks)

(b) a multiple of 4. (2 marks)

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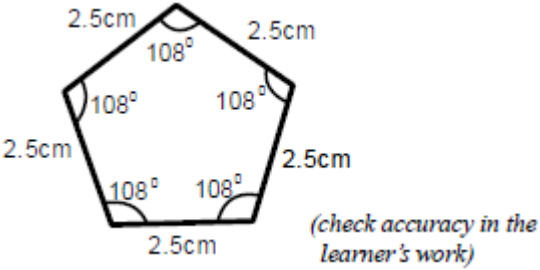
ANSWERS

Section A (Multiple Choice)

Qn	Answer	Qn	Answer	Qn	Answer	Qn	Answer	Qn	Answer
1	C	2	D	3	A	4	C	5	B
6	D	7	B	8	A	9	C	10	A
11	B	12	B	13	A	14	B	15	C
16	B	17	C	18	D	19	B	20	A

Section B (Structured)

Qn	Working / Explanation	Final Answer
21	$\log_{10} 10000 = 4$	4
22	$\frac{1}{2} * (\frac{10}{3} * \frac{2}{5}) +$ $\frac{43}{4} \div \frac{1}{4} = \frac{2}{3} +$ $\frac{19}{4} \times \frac{4}{1} = \frac{2}{3} + 19$	$19\frac{2}{3}$
23	$p \quad d = x.$ (Ratio proportion: $\frac{8}{4} = \frac{4}{8} ?$)	4 days
24	a) $P = 2(2x + 4 + x - 2)$ $< 40 \rightarrow 2(3x + 2) < 40$ b) $6x + 4 < 40 \rightarrow 6x < 36$	$p < 40$ $x < 6$
25	$2y + 3x = 12.$ At $y=0 \rightarrow$ $x=4$ (A= (4,0)), At $x=0$ $\rightarrow y=6$ (B= (0,6))	A(4,0), B(0,6) Accept any other working method
26	$I = \text{PRT}/100.$ $60000 \times$ $2 \times 10/100 = 12000.$ A $= 60000 + 12000$	Sh.72,000

27	$V = 25872 \text{ cm}^3 - 3234 \text{ cm}^3$	$22,638 \text{ cm}^3$
28	Interior angle = $180(n-2)/n$. For $n=5$: $180(3)/5 = 108^\circ$	108° 
29	a) Numbers between 30 and 40 = 11 b) Multiples of 4 = 32,36,40 Primes = 31,37 c) $P(4) = \frac{3}{11}$, $P(\text{prime}) = \frac{2}{11}$	Multiples=3, Primes=2

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